AdaptConfig

AI-Powered Integration Configuration Platform
for Enterprise Lending

Team Nucleolus

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FinSpark Hackathon · IIT Patna

The Problem

Indian lending platforms integrate with **8+ external APIs**:

- Credit Bureaus (CIBIL, Experian)
- Aadhaar eKYC Verification
- GST Verification Service
- Payment Gateways (Razorpay, UPI)
- Fraud Detection Engines
- SMS/Email Notification
- Account Aggregator (AA)

Pain Points

- Manual config takes **2–4 weeks** per API
- Field mapping errors cause **data loss**
- No testing before production
- Each API has different auth, schema, SLA
- **8 integrations = 16–32 weeks**

Estimated cost:

Our Solution

Upload API Spec $\xrightarrow{\text{AI}}$ **Auto-Config** $\xrightarrow{\text{Test}}$ **Deploy**

From **weeks** to **minutes**. Zero manual field mapping.

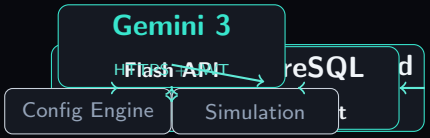
How It Works

1. Upload BRD / API spec (PDF, DOCX, YAML)
2. AI parses endpoints, fields, auth
3. Gemini 3 generates field mappings
4. Rule engine validates with confidence scores

Live Results (CIBIL Bureau v2)

Metric	Resul
Endpoints parsed	
Fields extracted	2
Auth schemes	2 (OAuth2 + API Key
Parse confidence	95%
Fields mapped	28/28 at 100%

System Architecture



Frontend
React 18, TypeScript

Tailwind, Recharts

Backend
FastAPI, SQLAlchemy

Pydantic v2, asyncio

AI / Infra
Gemini 3 Flash

Railway, Docker

Authentication & Multi-Tenancy

Security features:

- JWT access tokens (30 min) + refresh tokens (7 days)
- PBKDF2-HMAC-SHA256 password hashing with salt
- Per-user data isolation (unique tenant per account)
- Role-based access control (admin / editor / viewer)
- Auto-token refresh on expiry

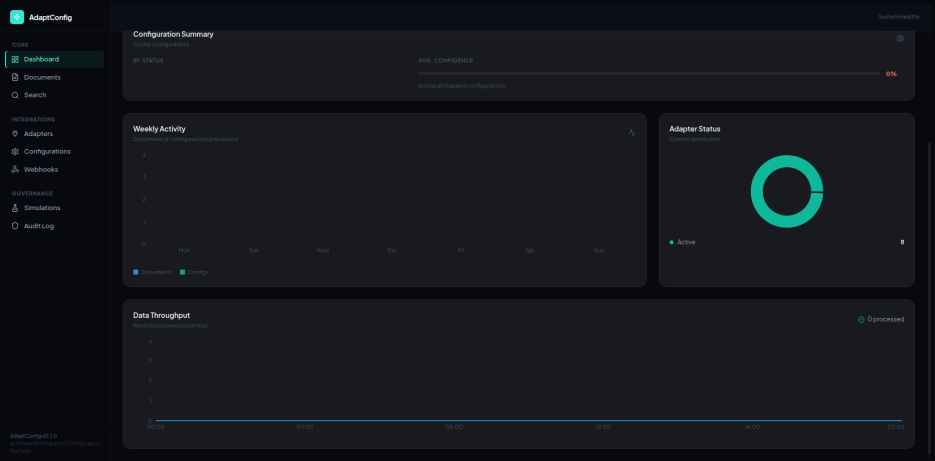
Each user sees only their own:

- Documents & parsed results

Registration & Login

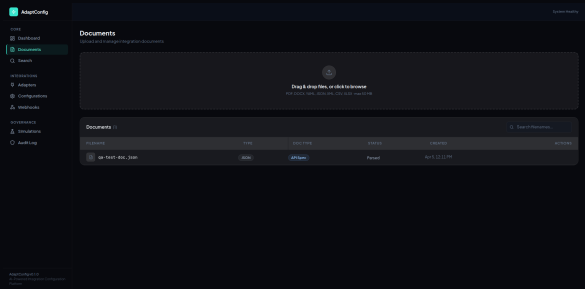
- Register with name, email, password
- Auto-login after registration
- Glassmorphism dark theme UI
- Sign out from sidebar

Dashboard



Real-time metrics: 8 adapters, configuration summary with confidence scores, weekly activity charts, adapter status distribution.

Step 1: Upload & Parse Document



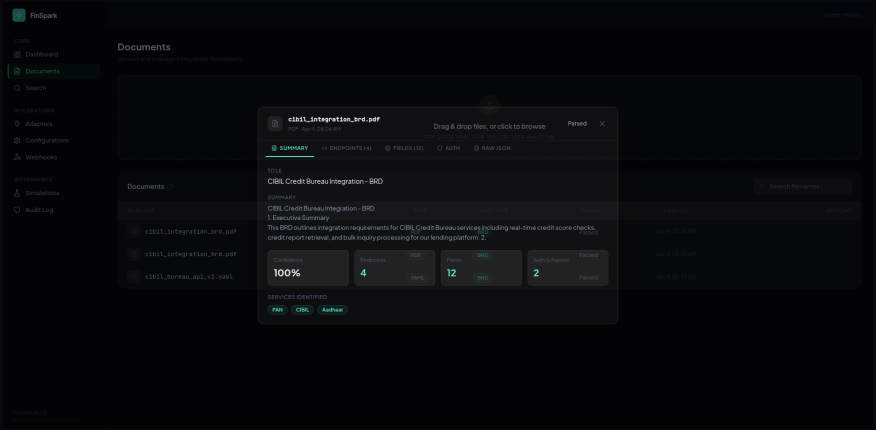
Supported formats:

- OpenAPI YAML / JSON
- Business Requirements (DOCX, PDF)
- Any API specification

Parser extracts:

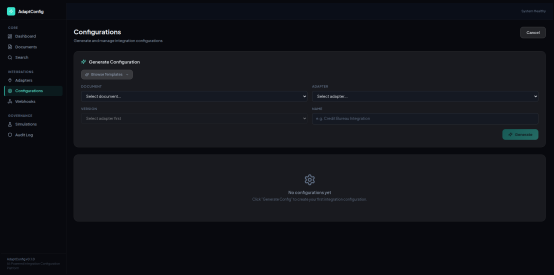
- API endpoints & HTTP methods
- Request/response fields with types
- Authentication schemes
- SLA requirements

Parsed Document Detail — 5-Tab Modal



Summary: Title, confidence, services identified · **Endpoints**: Path, method, description · **Fields**: 28 fields with types & required flags · **Auth**: OAuth2 + API Key · **Raw JSON**: Full parsed output

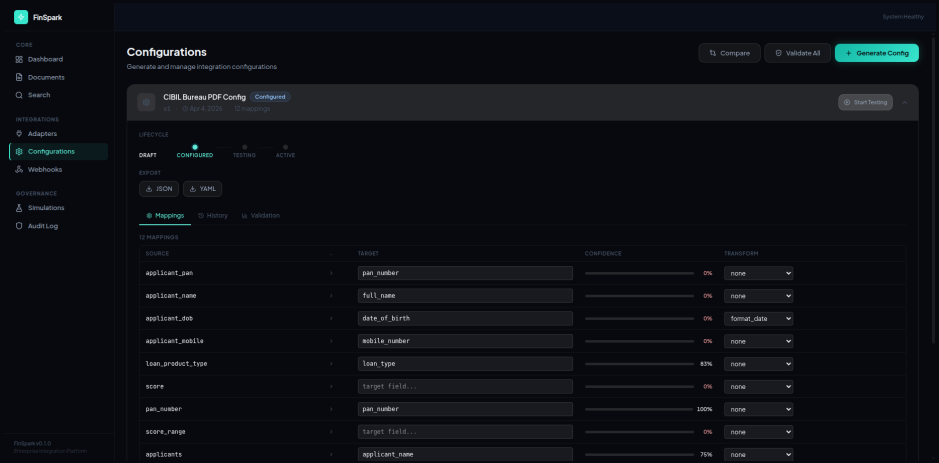
Step 2: AI-Powered Configuration Generation



Two-stage AI pipeline:

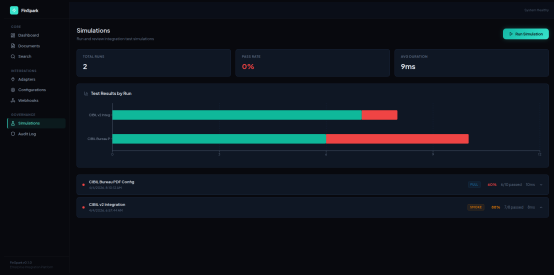
1. **Gemini 3 Flash** generates initial config from document entities + adapter schema
2. **Rule-based mapper** augments with:
   - Fuzzy field matching
   - Indian fintech synonym lookup (PAN, Aadhaar, GSTIN, IFSC)
   - Confidence scoring
   - Transform suggestions

Editable Field Mappings with Confidence



Lifecycle stepper (Draft → Configured → Validating → Testing → Active) · Editable target fields · Transform dropdowns · Color-coded confidence bars · Export

Step 3: Simulation & Testing

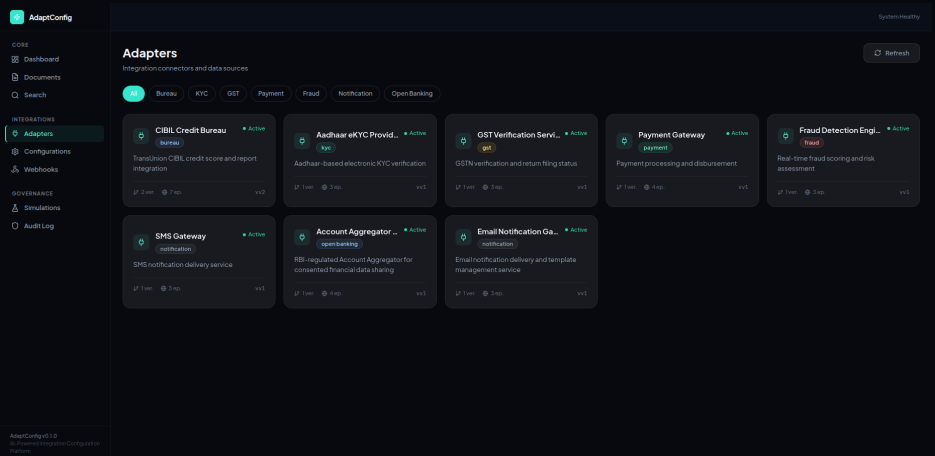


8-step validation pipeline:

1. Config structure validation
2. Field mapping coverage
3. Endpoint test: /scores
4. Endpoint test: /reports
5. Endpoint test: /batch/inquiries
6. Endpoint test: /consent/verify
7. Auth config validation
8. Hooks validation

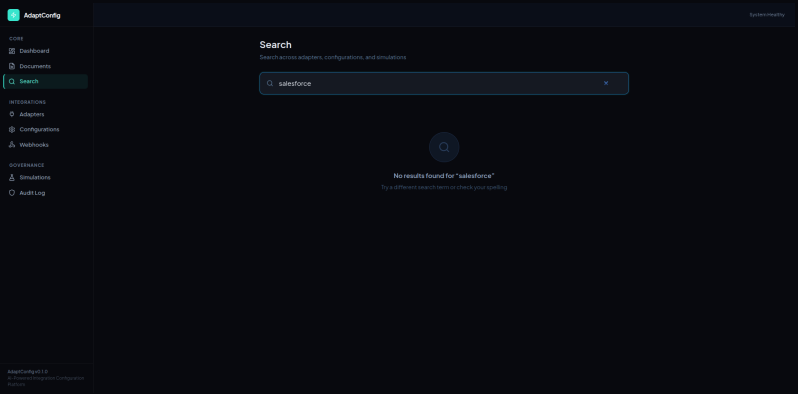
8/8 PASSED

Pre-Built Adapter Catalog



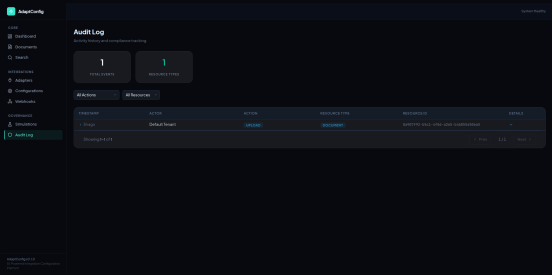
8 adapters: CIBIL (v1+v2), Aadhaar eKYC, GST, Payment Gateway, Fraud Detection, SMS, Account Aggregator (AA), Email · Category filtering · Version details with

Cross-Platform Search

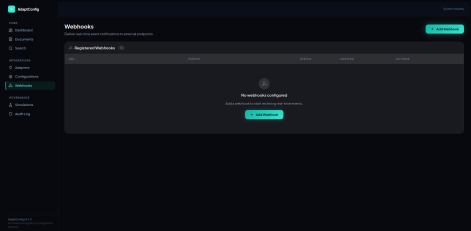


Fuzzy search across adapters, configurations, and simulations · Results grouped by type with relevance scores · Debounced real-time results

Governance: Audit Trail & Webhooks



Audit Log: Immutable trail of all actions with filters and pagination



Webhooks: Register URLs for event callbacks, test delivery, manage subscriptions

Security & Compliance

Authentication

- JWT Bearer tokens (HS256)
- PBKDF2 + salt password hashing
- Token refresh flow
- RBAC (admin / editor / viewer)

Data Protection

- Per-user tenant isolation
- PII masking (Aadhaar, PAN, phone)
- AES-256 Fernet encryption

Infrastructure

- CORS restricted to known origins
- CSP, HSTS, X-Frame headers
- Rate limiting (100 req/min)
- Structured JSON logging
- Graceful shutdown (30s drain)
- Alembic database migrations

Testing

- **899 automated tests**
- **82% code coverage**

Key Features Summary

Category	Features
Parsing	PDF, DOCX, YAML, JSON · endpoints, fields, auth extraction · confidence scoring
AI Config	Gemini 3 Flash · fuzzy field matching · Indian fintech synonyms · transform suggestions
Adapters	8 pre-built · auto-create from any API spec · category filtering · versioning
Simulation	8-step validation · mock responses per adapter · step-by-step results
Lifecycle	Draft → Configured → Validating → Testing → Active · rollback · history

Results & Metrics

Metric	Value
API Endpoints	34
Pre-built Adapters	8
Automated Tests	899
Code Coverage	82%
Frontend Pages	8
Lines of Code	∼15,000

Performance	Value
Parse confidence	95%
Mapping accuracy	100%
Simulation pass	8/8
Config gen time	< 5s
Doc parse time	< 1s
Test suite time	∼110s

	Manual	AdaptConfig	Improvement
Time per integration	2–4 weeks	< 5 minutes	∼500x faster

Live Demo

Live at:
https://adaptconfig-frontend-production.up.railway.app

1. Register / Login
2. [illegible]
3. [illegible]
4. [illegible]
5. [illegible]
6. [illegible]
7. Bifurcation Active

Thank You

AdaptConfig

AI-Powered Integration Configuration Platform

Live App	`https://adaptconfig-frontend-production.up.railway.app`
API Docs	`https://adaptconfig-api-production.up.railway.app/docs`
GitHub	`https://github.com/Akasxh/adaptconfig`

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